

# q. 11e

207765918 :577

ת. ד. ל. ג. א. ס. פ. :ל. י. נ.

$$g = \begin{cases} 2x & x < 0 \\ 1 & x \geq 0 \end{cases} \quad \text{פונקציה} \quad (1.1)$$

$$f(v) = \begin{cases} v^2 & v < 0 \\ 2v & v \geq 0 \end{cases}, \quad f(u) = \begin{cases} u^2 & u < 0 \\ 2u & u \geq 0 \end{cases}$$

ס. פ. :ל. י. נ.  $g(u)$ ,  $u \in \mathbb{R}$  ס. פ. :ל. י. נ.

$$f(v) \geq f(u) + g(u) \cdot (v - u) \quad : v \in \mathbb{R}$$

$$f(v) = \begin{cases} v^2 & v < 0 \\ 2v & v \geq 0 \end{cases}, \quad f(u) = u^2, \quad g(u) = 2u \quad \text{: } u < 0 \text{ נבחר פונקציה}$$

$$\begin{cases} v^2 & v < 0 \\ 2v & v \geq 0 \end{cases} \geq \underbrace{u^2}_{f(u)} + \underbrace{2u}_{g(u)}(v - u) \quad \text{: } u < 0 \text{ נבחר פונקציה}$$

$$\begin{aligned} u^2 + 2u(v - u) &= u^2 + 2uv - 2u^2 = 2uv - u^2 = v^2 - v^2 + 2uv - u^2 = \\ &= v^2 - (v^2 - 2uv + u^2) = v^2 - \underbrace{(v - u)^2}_{\geq 0} \leq v^2 = f(v) \end{aligned}$$

! 5k)

$$f(v) = 2v, g(u) = 2u \quad \text{! } \underline{v \geq 0} \text{ } \text{! } \underline{u \geq 0}$$

$$u' + 2uv - 2u^2 = \underbrace{2uv}_{(2uv \leq 0 \Leftarrow u < 0, v \geq 0)} - \underbrace{u^2}_{> 0} < 0 \leq 2v = f(v)$$

! u ≥ 0 } / > r - r)

! 5k) . g(u) = 2 f(u) = 2u, f(v) = 2v ! v ≥ 0 } / > r

$$f(u) + g(u)(v-u) = 2u + 2(v-u) = 2v \leq f(v)$$

||u|| - ||v||    ||v||    ||u|| →

! 5k) g(u) = 2, f(u) = 2u, f(v) = v^2 ! v < 0 } / > r

$$f(u) + g(u)(v-u) = 2u + 2(v-u) = 2v \leq 0 \leq v^2 = f(v)$$

v < 0 →    v^2 > 0

: u, v ∈ ℝ    √3    2    1, 1, 1, 1    2, 2, 2

g(u) ∈ ∂f(v)    ! f(v) ≥ f(u) + g(u) · (v-u)

$$x_0 = -1.5, \quad \eta = 0.15 \quad (1.5)$$

| $i$ | $x_i$                                                                        | $f(x_i) = \begin{cases} x & : x \geq 0 \\ x^2 & : x < 0 \end{cases}$ | $\frac{d}{dx} f(x_i) = g(x_i) = \begin{cases} 1 & : x_i \geq 0 \\ 2x_i & : x_i < 0 \end{cases}$ |
|-----|------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 0   | -1.5                                                                         | 2.25                                                                 | -3                                                                                              |
| 1   | $-1.5 - \frac{1}{4} \cdot (-3) = -\frac{3}{4}$                               | $\frac{9}{16}$                                                       | -1.5                                                                                            |
| 2   | $-\frac{3}{4} - \frac{1}{4} \cdot (-1.5) = -\frac{3}{8}$                     | $\left(\frac{-3}{8}\right)^2 = \frac{9}{64}$                         | $-\frac{6}{8}$                                                                                  |
| 3   | $-\frac{3}{8} - \frac{1}{4} \cdot \left(-\frac{6}{8}\right) = -\frac{1}{16}$ | $\left(\frac{-3}{16}\right)^2 = \frac{9}{256}$                       | $-\frac{6}{16}$                                                                                 |

רשימת  $x_i$  הנבחרת

הערות:  $x_i$  הוא הערך של  $x$  ברגע  $i$ .  
 הערה:  $x_i$  הוא הערך של  $x$  ברגע  $i$ .

$$x_0 = -1.5, \quad \eta = 1 \quad (1.4)$$

| $i$ | $x_i$                       | $f(x_i) = \begin{cases} x: x \geq 0 \\ x^2: x < 0 \end{cases}$ | $\frac{d}{dx} f(x_i) = g(x_i) = \begin{cases} 2: x_i \geq 0 \\ 1x_i: x_i < 0 \end{cases}$ |
|-----|-----------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 0   | -1.5                        | 2.25                                                           | -3                                                                                        |
| 1   | $-1.5 - 1 \cdot (-3) = 1.5$ | 3                                                              | 2                                                                                         |
| 2   | $1.5 - 1 \cdot 2 = -0.5$    | 0.25                                                           | -1                                                                                        |
| 3   | $-0.5 - 1(-1) = 0.5$        | 1                                                              | 2                                                                                         |
| 4   | $0.5 - 1 \cdot 2 = -1.5$    | 2.25                                                           | -3                                                                                        |

התהליך מתחיל בנקודה  $x_0$  ונבחר  $\eta = 1$  (הצעד המלא)

הנקודה הבאה היא  $x_1 = x_0 - \eta \cdot g(x_0) = -1.5 - 1 \cdot (-3) = 1.5$

הנקודה הבאה היא  $x_2 = x_1 - \eta \cdot g(x_1) = 1.5 - 1 \cdot 2 = -0.5$

הנקודה הבאה היא  $x_3 = x_2 - \eta \cdot g(x_2) = -0.5 - 1 \cdot (-1) = 0.5$

$$h(x) = 2x + x^3$$

(2.1) נתון, נחשב

$$h(0) = 0 = 0 = \gamma(0), \quad h(1) = 2 + 1 = 3 = \gamma(1) \quad ! \text{ ב"פ"ד}$$

$$h(2) = 2 \cdot 2 + 2^3 = 12 = \gamma(2)$$

$$h(0) = 0 \quad \text{נתון} \quad \text{נניח} \quad h(x) = \sigma x^2 + bx + c \quad \text{נ"ל} \quad (2.2)$$

$$h(x) = \sigma x^2 + bx \in C^0 \quad \text{נ"ל} \quad \text{נניח} \quad \sigma = 3, \quad b = 0$$

$$\begin{cases} h(1) = \sigma + b = 3 \\ h(2) = 4\sigma + b = 12 \end{cases} \quad \text{נניח} \quad \text{נניח} \quad \sigma = 3, \quad b = 0$$

נניח

$$L(w) = \frac{1}{n} \|Xw - \gamma\|_2^2 \quad (2.3)$$

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 4 & 4 & 8 \end{bmatrix}, \quad w = \begin{pmatrix} 0 \\ 2 \\ 0 \\ 1 \end{pmatrix}, \quad \gamma = \begin{pmatrix} 0 \\ 3 \\ 12 \end{pmatrix}, \quad n = 3$$

$$\Rightarrow L(w) = \frac{1}{3} \left\| \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 4 & 4 & 8 \end{bmatrix} \begin{pmatrix} 0 \\ 2 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \\ 12 \end{pmatrix} \right\|_2^2 =$$

$$= \frac{1}{3} \left\| \begin{pmatrix} 0 \\ 3 \\ 12 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \\ 12 \end{pmatrix} \right\|_2^2 = 0$$

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \quad W = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}, \gamma = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} \quad : (2.1) \quad \text{see (1.1)}$$

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$$L(W) = \frac{1}{3} \left\| \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} \right\|_2 = \frac{1}{3} \left\| \begin{pmatrix} 0 \\ 3 \\ 7 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} \right\|_2 = 0$$

$\gamma = 6x - 1 \Rightarrow W = \begin{pmatrix} -1 \\ 6 \end{pmatrix} \quad (2.9)$

$$\Rightarrow L(W) = \underbrace{\left\| \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} -1 \\ 6 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\|_2}_1 = \underbrace{\left\| \begin{pmatrix} -1 \\ 5 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\|_2}_2 = \underbrace{\left\| \begin{pmatrix} -1 \\ 4 \end{pmatrix} \right\|_2}_3 = 2$$

:(2.1) Mar (2.5)

$$L_\lambda(W) = L(W) + \lambda \|W\|_1^2 = \lambda \left\| \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} \right\|_1^2 = 5\lambda$$

see (1.1) Mar

$$L_\lambda(W) = L(W) + \lambda \|W\|_1^2 = \lambda \left\| \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} \right\|_1^2 = 9\lambda$$

see (1.4) Mar

$$L_\lambda(W) = L(W) + \lambda \|W\|_1^2 = 2 + \lambda \left\| \begin{pmatrix} -1 \\ 6 \end{pmatrix} \right\|_1^2 = 2 + 37\lambda$$

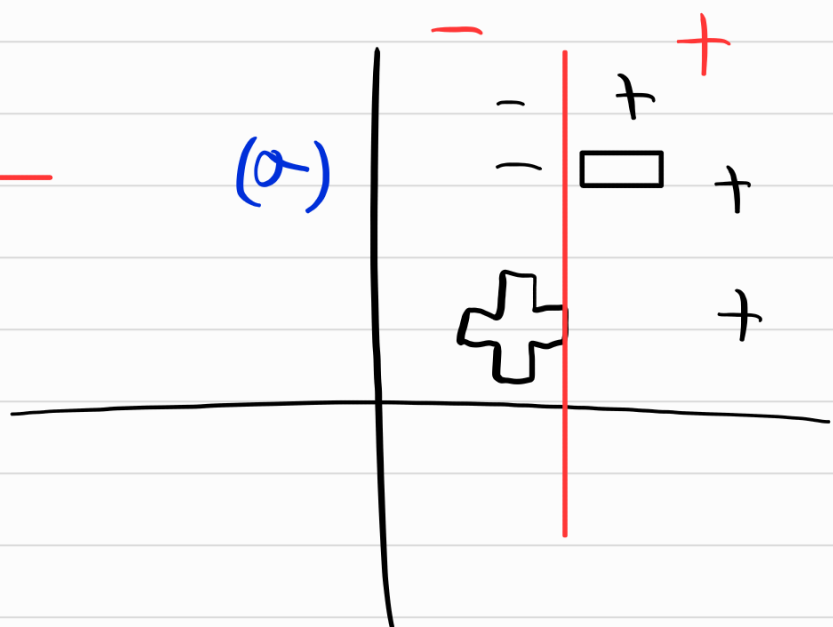
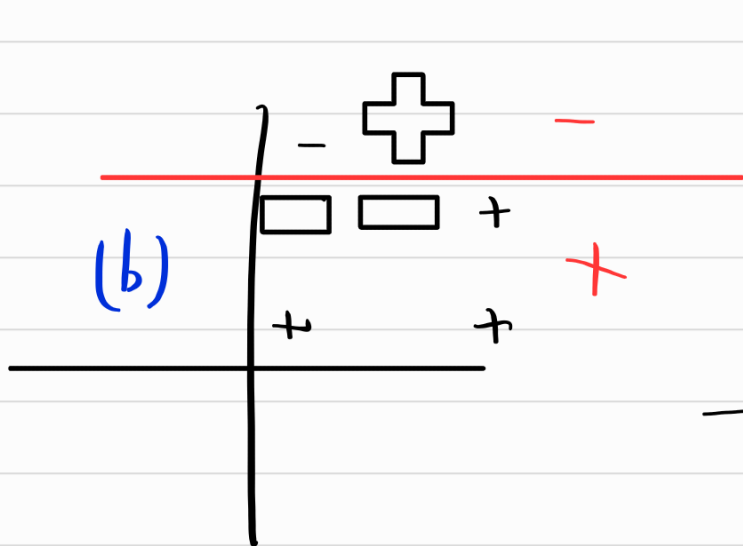
(7.6) קויד'א כ' י'ד'א וק'א'א'א'א'א'א'א'א'א'א'א'א'א'א'א'א'א'א'א'א'א'א'א'A

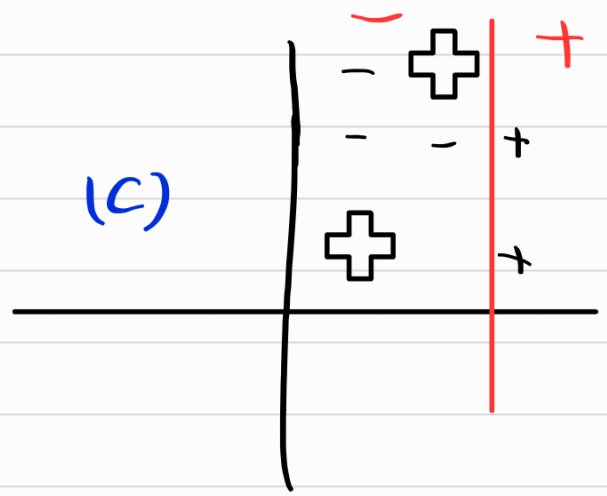
$$\sqrt{s} \rho \quad g_\lambda > s_\lambda \quad , \quad \lambda > 0 \quad \text{e} \quad |'' \cdot \cdot \cdot \cdot | \quad L_\lambda(w) = g_\lambda \quad \rho \cdot \lambda |'' \cdot \cdot \cdot |$$
$$\lambda \dim(W) \sim \lambda \|W\|_1 \quad \text{for } n, p \rightarrow \infty \quad (2.7)$$

$L_\lambda(u) = \sum_{\lambda} p(\lambda) u^\lambda$

[illegible]

$\sigma, b, c$  :  $\rho, \gamma$  :  $\gamma$  : AdaBoost  $\gamma$





! 4 2 1 2 e

$$E = \{x_i \mid i \in [m], \gamma_i \neq h_t(x_i)\} \quad ! 1 1 1 1$$

$$\sum_{i=1}^m D_i^{t+1} \mathbb{I}[\gamma_i \neq h_t(x_i)] = \sum_{x_i \in E} D_i^{t+1} = \sum_{x_i \in E} \frac{D_i^t \exp(-\alpha_t \gamma_i h_t(x_i))}{\sum_j D_j^t \exp(-\alpha_t \gamma_j h_t(x_i))} \quad ! 1 1 1$$

$$10/12 \quad x_i \in E \quad \text{is } \gamma_i \cdot h_t(x_i) = -1 \quad \text{and } \gamma_i \in \{-1, 1\} \quad \text{and } \gamma_i \neq h_t(x_i)$$

$$\text{is } \gamma_i \cdot h_t(x_i) = -1 \cdot 1 = -1 \quad \text{and } \gamma_i \in \{-1, 1\} \quad \text{and } \gamma_i \neq h_t(x_i)$$

$$= \sum_{x_i \in E} \frac{D_i^t \exp(-\alpha_t \cdot (-1))}{\sum_j D_j^t \exp(-\alpha_t \gamma_j h_t(x_i))} = \sum_{x_i \in E} \frac{D_i^t \exp(\alpha_t)}{\sum_j D_j^t \exp(-\alpha_t \gamma_j h_t(x_i))}$$

$$\alpha_t = \frac{1}{\epsilon_t} \log\left(\frac{1}{\epsilon_t} - 1\right)$$

$$: \mu(\lambda) \rightarrow \text{probability}$$

$$\exp\left[\frac{1}{\epsilon_t} \log\left(\frac{1}{\epsilon_t} - 1\right)\right] = \exp\left[\log\sqrt{\frac{1}{\epsilon_t} - 1}\right] = \sqrt{\frac{1}{\epsilon_t} - 1}$$

$$\Rightarrow \frac{\sum_{x_i \in E} D_i^t \exp(\alpha_t)}{\sum_{j \in D_j^t \exp(-\alpha_t \gamma_j h(x_j))}} = \frac{\sum_{x_i \in E} D_i^t \sqrt{\frac{1}{\epsilon_t} - 1}}{\sum_{j \in D_j^t \exp(-\alpha_t \gamma_j h(x_j))}}$$

$$\sum_{j \in D_j^t \exp(-\alpha_t \gamma_j h(x_j))} = \sum_{x_i \in E} D_i^t \exp(\alpha_t) + \sum_{x_i \notin E} D_i^t \exp(-\alpha_t)$$

$$\sum_{j \in D_j^t \exp(-\alpha_t \gamma_j h(x_j))} = \sum_{x_i \in E} D_i^t \exp(\alpha_t) + \sum_{x_i \notin E} D_i^t \exp(-\alpha_t) =$$

$$= \sum_{x_i \in E} D_i^t \sqrt{\frac{1}{\epsilon_t} - 1} + \sum_{x_i \notin E} D_i^t \sqrt{\frac{1}{\epsilon_t} - 1}$$

$$\frac{\sum_{x_i \in E} D_i^t \sqrt{\frac{1}{\epsilon_t} - 1}}{\sum_{j \in D_j^t \exp(-\alpha_t \gamma_j h(x_j))}} = \frac{\sum_{x_i \in E} D_i^t \sqrt{\frac{1}{\epsilon_t} - 1}}{\sum_{x_i \in E} D_i^t \sqrt{\frac{1}{\epsilon_t} - 1} + \sum_{x_i \notin E} D_i^t \sqrt{\frac{1}{\epsilon_t} - 1}} =$$

$$= \frac{\sum_{x_i \in E} D_i^t \sqrt{1 - \epsilon_t}}{\sum_{x_i \in E} D_i^t \sqrt{1 - \epsilon_t} + \sum_{x_i \notin E} D_i^t \sqrt{\frac{\epsilon_t}{1 - \epsilon_t}}} = \frac{\sqrt{1 - \epsilon_t} \sum_{x_i \in E} D_i^t}{\sqrt{1 - \epsilon_t} \sum_{x_i \in E} D_i^t + \sqrt{\frac{\epsilon_t}{1 - \epsilon_t}} \sum_{x_i \notin E} D_i^t}$$

$$\epsilon_t = \frac{\sum_{x_i \in E} D_i^t}{\sum_{x_i \in E} D_i^t + \sum_{x_i \notin E} D_i^t} \Rightarrow \sum_{x_i \notin E} D_i^t = \frac{\sum_{x_i \in E} D_i^t}{\epsilon_t} - \sum_{x_i \in E} D_i^t$$



הערה: כל המידע הנ"ל הוא למטרות לימודיות בלבד.

$\gamma \in \Gamma$      $\gamma \in \Gamma$      $\gamma \in \Gamma$      $\gamma \in \Gamma$      $\gamma \in \Gamma$

$$N-2 \quad B_{\sim 0} \quad R \quad A \quad B$$
$$\hat{\gamma}_i \text{ wegen } Z = \{x_i \mid x_i : \hat{\gamma}_i := \gamma_i\} : \mu_0 . \quad \text{Nr}$$
[illegible]

$$w_1 = w_0 + \sum_{i=1}^n \eta y_i x_i = \eta \left[ \sum_{i=1}^n y_i x_i \right] \quad \text{...}$$

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$$W_2 = y_1 + \sum_{z_2} \eta y_i x_i = \eta \left[ \sum_{z_1} y_i x_i \right] + \eta \left[ \sum_{z_2} y_i x_i \right] =$$

$$= \eta \left[ \sum_{z_1} y_i x_i + \sum_{z_3} y_i x_i \right]$$

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$$W_n = W_{n-1} + \sum_{i=1}^N \eta y_i x_i = \eta \sum_{j=1}^N \sum_{i=1}^N y_i x_j$$

$$\hat{y}_i = \text{sigmoid}(w^T x_i)$$

ה'תשנ"א, י"ב באב

(כאשר  $w$  וקטור כשני (מנגי' בכיוון  $w$ ) וקטן  $\alpha$ )

למה? כי כיוון  $w$  קטן  $\alpha$  באיטיות. רק באיטיות

ה  $N$  בנקודת  $n$  נקודת  $w$  (המק"ב  $(x_i^T w) = \gamma_i$ ,  $\gamma_i \in [0, 1]$ )

ה  $\mu$  נקודת  $w$  הנוקדית.  $\gamma_i$ ,  $\gamma_i \in [0, 1]$

מכאן  $\gamma_i \in [0, 1]$ ,  $w$  וידי  $\gamma_i$ ,  $\gamma_i \in [0, 1]$  באיטיות

כאשר